OBJECT DETECTION USING YOLO-V8

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Abstract

The growing need for automated object detection and recognition in visual data has become crucial in various domains, including surveillance, autonomous vehicles, and image analysis. Traditional methods often struggle with speed and accuracy, making real-time applications challenging. The main objective of this experiment is to make the computers identify and detect objects in the visual data. The algorithm is used to detect the object with classification and localization based on visual inputs. This project uses YOLO(You Only Look Once) algorithm to detect the objects. YOLO v8 is known for its speedy and accurate object detection. This idea is used to identify multiple objects in images, videos, and live webcams. This algorithm adapts to various scenarios such as image analysis to dynamic object tracking.

Keywords *Object detection, YOLO, Dynamic object tracking, Visual data, Classification, Recognition.*

I.INTRODUCTION

In this fast-moving world, everything is computerized and object detection plays a crucial role in everyone's life. We are all using mobile phones and it has the feature to unlock mobile using face recognition. The unlock feature follows the computer vision technique as face recognition is an application of object detection. In security systems, CCTV cameras adapt object detection mechanisms to identify unusual activities. Object detection can be done by various algorithms and methods, here we have used YOLOV8 (You Only Look Once Version 8) an effective object detection algorithm. Object detection in the virtual data is mainly based on image classification, localization, and detection. YOLOV8 algorithm classifies the object using a convolutional neural network locates the objects and finally detects that object with some probability scores with minimum background errors. Currently we are using YOLOV8, but YOLO has many versions such as YOLOV1, YOLOV2,YOLOV3,YOLOV4,YOLOV5,YOLOV6 and YOLOv7

II.EXISTING SYSTEM

First, most of the object detection system relies on manual labor which results in wrong outcomes and might take a longer time to detect. Most of the object detection system produces an inaccurate result and cause delays. This may not be suitable for emergency conditions like surveillance cameras and auto-driving vehicles.

III.PROPOSED SYSTEM

It can process three various kinds of data including live feeds, videos, and images. The capacity to recognize 85 distinct classifications. It uses both pre-trained models and custom models and offers a straightforward interface.

IV.LITERATURE SURVEY

D.G. Lowe's paper on Scale-Invariant Feature Transform (SIFT) [4], which presented a technique for identifying and characterizing local features in images, is credited with initiating early work in object identification. This paved the way for later developments by allowing computers to locate things within a picture with reliability.

Papageorgiou, Oren, and Poggio (1998) [9] presented a technique that makes use of wavelet characteristics and support vector machines (SVMs) to recognize objects in images. This method was one of the first to show reliable object detection in still photos, and it provided a systematic strategy that impacted other studies.

The introduction of deep learning techniques resulted in significant advancements. 2015 saw the release of Girshick's Fast R-CNN [10], a technique that greatly increased object detection speed and accuracy. Remarkably, Fast R-CNN achieved real-time object identification performance by proposing regions of interest and then classifying these regions using region proposal networks.

YOLO underwent multiple iterations in its evolution, each one building on the one before it.

- improved model design for feature extraction and fusion with a more effective neck and backbone.
- enhanced anchor-free detection head, which makes anchor box configurations simpler.
- sophisticated training methods and data augmentation techniques to improve the performance and robustness of the model.

When we contrast these new models with YOLOv8, we discover:

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- For applications that demand fast decision-making, YOLOv8 continues to have an advantage in speed and real-time performance.
- EfficientDet is a computationally and parameter count-efficient solution that may be deployed on devices with limited resources.
- With its transformer-based methodology, DETR offers a novel viewpoint and is advantageous in managing intricate interactions between items, though it might need more processing power.

Models like EfficientDet [14] and DETR [15] have been developed as a result of recent advances in object detection.

V. METHODOLOGY

The user can give the inputs in any format such as image, video, or live. The program works at the speed of 150 frames per second. The user can give one image or a video clip (either recorded or live video captured using a webcam that is attached to the laptop/system). It predicts the bounding boxes and confidence score with the class probabilities.

A. YOLOv8 Algorithm

YOLO(You Only Look Once) is an object detection technique. According to the researchers, the YOLO algorithm architecture is extraordinarily faster than the other object detection algorithms [2,14,15]. While the network's smaller variant, Fast YOLO, processes an amazing 150 frames per second while still attaining double the mAP of other real-time detectors, the base YOLO model processes images in real time at a rate of 45 frames per second. As this algorithm generalizes from natural images to other domains such as artwork, it performs better than other detection approaches such as DPM and R-CNN.

B. CNN(Convolutional Neural Network)

The CNN(Convolutional Neural Network) serves as a backbone of object detection with the new architecture CSPDarknet [14-16]. CNN is used to extract relevant features from the images such as edges, textures, and shapes for understanding the content of the image and It generates feature maps. The feature maps are then processed through the feature pyramid network, which detects objects of different sizes in an image by creating a "pyramid". and adjusts the resolution of the feature maps. this helps yolov8 suitable for various applications.

C. Requirements

The technology used in this project is Python. The minimal hardware requirement for the proposed system is 4 GB RAM, Standard SSD/HDD storage, and Windows 11 operating system.

VI. ARCHITECTURE DIAGRAM

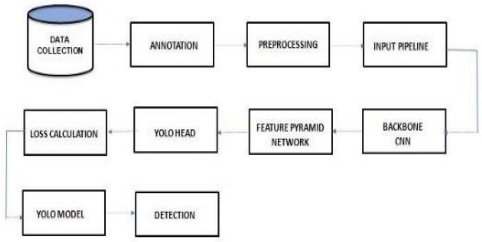


Fig. 1. Architecture diagram

VII. WORKFLOW DIAGRAM

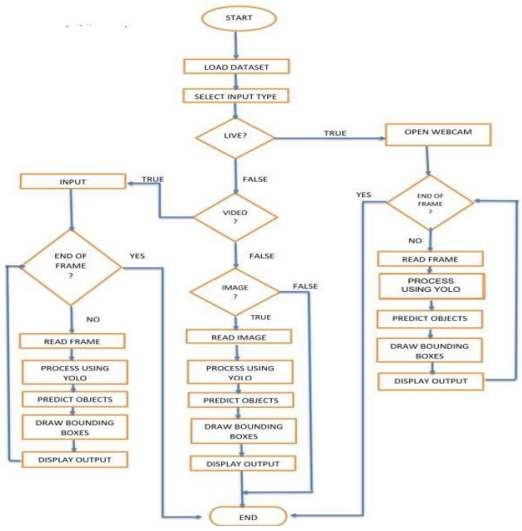


Fig. 2. Workflow of the object detection process.

VIII. DATASET

We collected images for the datasets in real-time, like our college identity cards, tags, and faces of people. some of the images were downloaded directly from Google Images to our needs.

A. Preprocessing

All images are imported. Roboflow was used to carry out the annotating procedure. Bounding boxes are drawn around noteworthy items in each image, and these boxes are labeled with the class of the object. Precise annotations are essential for the model to understand the precise locations and object classes. Inadequate annotations might cause a model to miss things or identify them wrongly, which lowers the model's overall effectiveness and dependability. The annotated dataset is preprocessed with auto-orientation, resize (650x650), and noise (+0.1). By correcting rotations, auto-orientation makes sure that every image is orientated appropriately. By ensuring uniform input dimensions by standardization of the image size, memory consumption, and computational complexity are decreased. By adding noise, the model can become more resilient to changes and more capable of generalizing to new, untested data.

Then the dataset is downloaded in YOLOv8-oriented bounding boxes format. We also implemented a pre-trained model (YOLOV8n.pt) downloaded directly from the Ultralytics GitHub page which contains 80 classes.

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B. Training

We collected the dataset of images and labels were given as input for training then they were processed in the backbone CNN. The backbone CNN extracts the abstract information of the image like edges, textures, shapes, and structures [2,3]. It generates feature maps. The feature maps are then processed through the feature pyramid network. FPN helps YOLOv8 detect objects of different sizes and It adjusts the resolution of the feature maps[14-16]. Multiple Yolo heads process the features and predict the bounding box for objects by continuously predicting the parameters in each grid. It calculates the confidence score that indicates the probability that the box contains the object. It also predicts the class probabilities. The predictions are compared to the original annotations using a custom loss function. The error is given through the network and the model parameters are updated to minimize the loss function. After training, The model is created, and that model is used for object detection.

IX. RESULTS

The following figures are the results produced during training.

A. Key Metrics

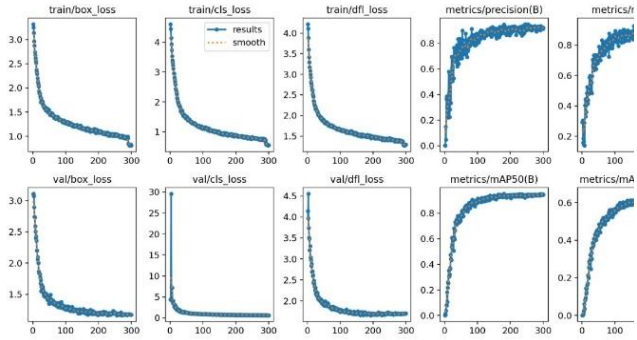


Fig. 3. The key metrics are generated for training 300 epochs. The precision shows the accuracy of the detected objects, Recall shows the ability of the model to identify the objects, and the mAP (Mean Average Precision) provides a comprehensive evaluation of the model's performance.

B. Confusion matrix

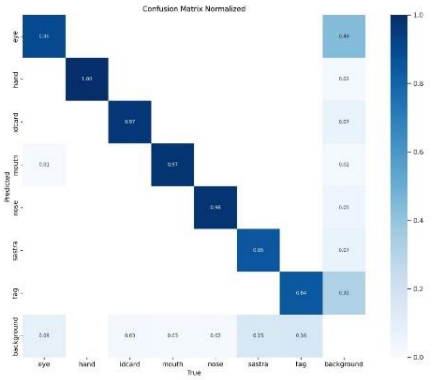


Fig. 4. The confusion matrix shows the counts of true positives, true negatives, false positives, and false negatives for the classes eye, hand, ID card, mouth, nose, sastra, and tag.

C. F1- confidence curve

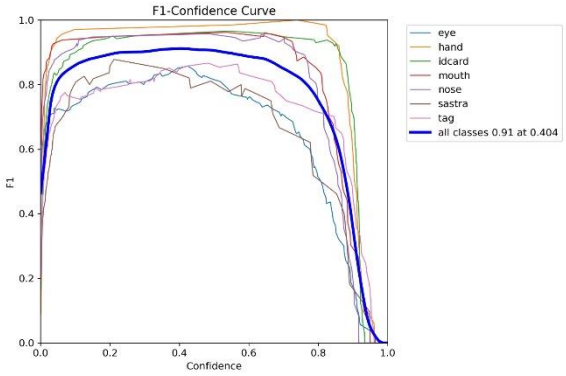


Fig. 5. The f1-confidence curve plots the f1 score against confidence thresholds. The higher score indicates better performance.\

D. Precision-recall curve

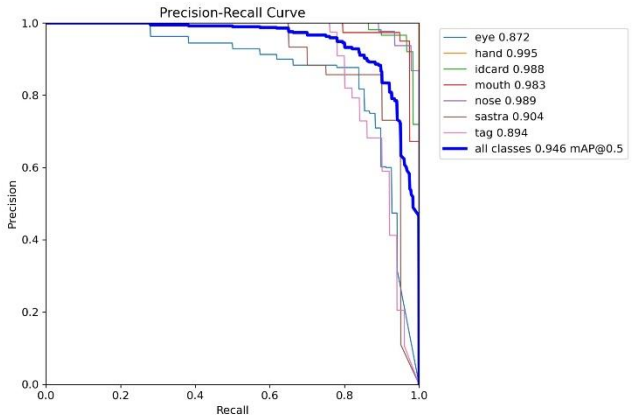


Fig. 6. The Precision quantifies the proportion of true positives among all positive predictions. Recall calculates the proportion of true positives among all actual positives.

X. COMPARISON

TABLE I. YOLOV8 VS OTHER ALGORITHMS

Algorithms	Key metrics		
	Precision	***Recall***	***mAP***
Yolov8	0.85	0.80	0.82
Yolov7	0.83	0.78	0.80
FasterR-CNN	0.86	0.83	0.84
SSD	0.80	0.75	0.78
EfficientDet	0.85	0.81	0.83

XI. OUTPUT

Fig. 5. is downloaded from Google Images, fig.6. is recorded video in real-time, and Fig. 7. is a live webcam. The output for image, video, and live are shown in the following figures.

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Fig. 7. Output of image

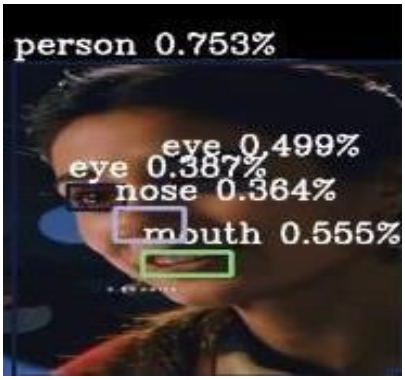


Fig. 8. Output of video



Fig. 9. Output of live

XII. Conclusion and Future Enhancements

The experiment focuses on enabling computers to swiftly and accurately identify objects in visual data. It utilizes the YOLOv8 algorithm, known for its speed and precision in object detection. Implemented in Python, the system is user-friendly and adaptable to various scenarios, from image analysis to dynamic object tracking. Future enhancements could include improving accuracy, optimizing speed, adding advanced features, ensuring scalability, and enhancing the user interface for a better experience.

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